

ALIAH UNIVERSITY

B. Tech. Semester:IV, Civil Engineering

End Sem. Examination, June-2022

Subject Code &Name: CENUGPC09, Engineering Geology

Time: 3 Hours

Total Marks: 80

Instruction: (1) Figures to the right indicate full marks.
(2) Assume suitable data if required.
(3) Attempt any five.

Que. – 1 Give the detailed account of geological work of seas and oceans.

16×5=80

Que. – 2 What do you mean by Lithosphere? What are the components of lithosphere discuss in detail.

Que. – 3 Discuss the following physical properties of minerals with example
Colour, lustre, fracture, streak, cleavage

Que. – 4 What is fault? Explain the parts and classification of fault in detail with neat sketches.

Que. – 5 Explain different geological parameters which are require for site selection of Dam.

Que. – 6 Discuss in detail the different shape of deposition made by river with neat sketches.

END OF PAPER

ALIAH UNIVERSITY
End Semester Examination-2021-22

B. Tech (3rd Semester)

Department of Civil Engineering

Subject Code: CENUGPC06 Subject Name: Fluid Mechanics & Hydraulics

Time: 3 Hr

Maximum Marks= 80

Note: Attempt any 5 questions. All questions carry equal marks.

1. The velocity potential function (ϕ) is given by an expression
$$\phi = -(XY^3/3) - X^2 + (X^3Y/3) + Y^2$$

(a) Find velocity component in X and Y direction (b) Velocity at point (5,2)
(c) Show that the potential function satisfies Laplace equation of flow
2. A circular plate 3.0 m diameter is immersed in water in such a way that its greatest and least depth below the free surface is 4 m and 1.5 m respectively. Determine the total pressure on one face of the plate and position of the centre of pressure
3. What do you mean by buoyancy and centre of buoyancy? Discuss the condition of equilibrium of a floating and submerged body.
4. A 45° reducing bend is connected in a pipeline, the diameter at the inlet and outlet of the bend being 600 mm and 300 mm respectively. Find the force exerted by water on the bend if the intensity of pressure at inlet to bend is 8.829 N/cm² and rate of flow is 600 litres/s.
5. Find the discharge of water flowing through a pipe 30 cm diameter placed in an inclined position where a venturimeter is inserted, having a throat diameter of 15 cm. The difference of pressure between the main and throat is measured by a liquid of specific gravity 0.6 in an inverted U-tube which gives a reading of 30cm. The loss of head between the main and throat is 0.2 times the kinetic head of the pipe.
6. Calculate the discharge through a pipe of diameter 20 cm and length 50 m when one end of the pipe is connected to a tank and other end of the pipe is open to the atmosphere. The pipe is horizontal and the height of water in the tank is 4 m above the centre of the pipe. Consider all minor loss and take $f=0.009$ in the formula $h_f = 4flv^3/2gd$. Also draw the hydraulic gradient line and total energy line.

Subject-Environmental Science
Course Code- UCCUGMC02
4th Sem

Full marks-80

Time-3 Hours

Group – A

Give tick (✓) any 20 questions from the following.

20 x 1 = 20

1. What was the name of the primitive super continent of the earth?
a. Asia b. Gondwanaland c. Pangea d. Panthalassa
2. The nearest star of the Earth is
a. Proxima centauri b. Sun c. Jupiter d. none of these
3. The word Ecology is used first time by
a. AG Tansley b. Ernst Haeckel c. Einstein d. Robert Hooke
4.% of the sun radiation reflected back to space. It is called albedo.
a. 50% b. 64% c. 82% d. 34%
5. From which Nebula our solar system was originated?
a. Solar Nebula b. Solar gas c. Orion Nebula d. Rosette Nebula
6. What is the age of our Universe?
a. 15.5 billion years b. 13.8 billion years c. 13.6 billion years d. 4.5 billion years
7. Which pyramid is always upright ?
a. Number pyramid b. Biomass pyramid c. Energy pyramid d. none of these
8. One non conventional sources of energy is
a. Coal b. Nuclear energy c. Solar energy d. Fossil fuel
9. In troposphere temperature always..... with increasing height.
a. Increases b. Decreases c. stable d. none of these.
10. One example of primary consumer is
a. Lion b. Rabbit c. Green plant d. Fox
11. The full form of DDT is
a. Di nitro di chloro toluene
b. diphenyl Dichloro toluene
c. dichloro diphenyl tri chloro ethane
d. none of these
12. How many species are present in the world, according to Camilo Mora (2011)?
a. 8.7 million b. 50 million c. 8.4 million d. 15.5 million
13. One of the main groundwater pollutant is
a. carbon dioxide b. Arsenic c. CFC d. none of these.
14. The full form of the UNDP is
a. United Nations Development Programme
b. United Nations Departmental Programme
c. United Nations Disaster Programme

d. United Nations Cultural Programme

15. The main culprit gas of Ozone layer depletion is-

- a. PFC b. CFC c. carbon dioxide d. water vapour

16. One example of water borne disease is-

- a. Pneumonia b. Cholera c. AIDS d. common cold

17. The Chipko movement started in

- a. Kerala b. West Bengal c. Western Ghat d. Garhwal Himalayan region

18. One of the biggest sources of SO₂ in the atmosphere is

- a. thermal power plants b. moving vehicles c. ocean d. green plants.

19. Ozone Layer protect us from harmful

- a. ultraviolet rays b. infrared rays c. visible rays d. none of these

20. In an ecosystem autotrophs are

- a. Animal b. Green plants c. Microbes d. Mammals.

21. The World Environment Day is celebrated on

- a. 5th June b. 22 April c. 5th September d. 25th December

22. Electrostatic Precipitator is used to control over

- a. Air Pollution b. Noise pollution c. Water pollution d. Soil pollution

23. Oxygen present in the atmosphere is

- a. 11% b. 21% c. 31% d. 41%

Group- B

Answer any 6 questions from the following.

1. What is the Environment ? Describe briefly about the atmospheric layers. 2+8

2. What is Ecosystem ? Discuss briefly about the food pyramid and different types of food pyramids in an ecosystem. 2+8

3. What is a biogeochemical cycle ? Describe the Hydrological cycle. 3+7

4. Define water scarcity. Mention the reasons of water scarcity in the world. What is eutrophication ? 2+5+3

5. What is an air pollutant ? Classify the air pollutants with proper examples. What is acid rain ? 2+7+3

6. What do you mean by Greenhouse effect ? What are the main Greenhouse gases in the atmosphere ? Write the impacts of global warming. 3+2+5

7. Write a notes on Silent Valley movement. 10

8. Write any two short notes from the following- 5X2=10

- A) Bhopal gas disaster
- B) Minamata disease
- C) Grassland ecosystem
- D) Biopiracy

Aliah University

End Semester Examination, June 2022,

Course: B. Tech. (Civil Engineering)

Year & Semester: 2nd Yr. (IV Sem.)

Subject: Design of RCC Structures

Subject Code: CENUGPC08

Full Marks: 80

Time: 3hrs.

Answer any four questions. All questions carry equal marks. IS code is allowed. Assume suitable data if required.

1. (a) Explain the principles of design philosophies used in RCC structures.
(b) Explain balanced, under reinforced and over reinforced sections.
[10+10=20]

2. Design a simply supported rectangular beam having 5 m effective span. It is subjected to a live load of $(15+X)$ kN/m (where X = last one digit of your Roll Number). Also design for shear reinforcement. Assume suitable data.
[20]

3. (a) Where doubly reinforced section is designed? Explain the design concept.
(b) Design a doubly reinforced section for a rectangular beam, a factored bending moment of 160 kN-m is acting at mid span where size of the beam is limited to 300 mm $\times$ 450 mm. Use M20 mix and Fe 415 grade steel. (Take $f_{sc} = 353$ MPa for $d'/d = 0.1$)
[10+10=20]

4. Design a simply supported roof slab for a room 8 m $\times$ 3.5 m clear in size if the superimposed load is 4 kN/m². Use M20 mix and Fe 415 grade steel. Do proper detailing.
[20]

5. Design a short square column to carry an axial load of $(1000+10X)$ kN (where X = last two digits of your Roll No.) kN using Fe 415 grade steel M25 concrete mix. Do proper detailing.
[20]

ALIAH UNIVERSITY
Paper Name: BIOLOGY FOR ENGINEERS
Paper Code: BIOUGBS01
2022

Marks-80

Time-3Hrs

A. Answer the following question with select the correct options.

1×45=45

1. The canal system is characteristic feature of
(A) Arthropods
(B) Mollusca
(C) sponges
(D) Echinoderms
2. Cistron is
(A) The coding sequence of DNA
(B) The functional unit of DNA molecule that codes for a particular gene product
(C) Intervening non coding sequence of DNA
(D) The sequences which are removed during RNA splicing.
3. This is the largest phylum of Animal on the earth.
(A) Mollusca
(B) Amphibia
(C) Arthropoda
(D) Aves
4. The jawless vertebrate is
(A) crocodile
(B) zoris
(C) *Hyla*
(D) *Petromyzon*
5. Ribosomes are made up of
(A) RNA only
(B) RNA and Proteins
(C) RNA, DNA and Proteins
(D) nucleic acids, proteins and lipids
6. Which of the following snake is non-poisonous?
(A) cobra
(B) krait
(C) viper
(D) python
7. Which of the following is the function of lysosomes?
(A) autophagy
(B) autolysis
(C) digestion
(D) all of the above
8. Phylum of *Taenia Solium* is
(A) Aschelminthes
(B) Annelids
(C) Platyhelminthes
(D) mollusca
9. Where is DNA present in the eukaryotic cells?
(A) Inside the nucleus
(B) With other cellular contents
(C) Inside the ribosomes
(D) Not present
10. Which of the following nitrogenous base is not present in DNA?
(A) Thymine
(B) Adenine
(C) Guanine
(D) Uracil
11. The bases are held together in a DNA double helix by hydrogen bonds. These bonds are
(A) Ionic bonds
(B) Covalent bonds
(C) Non-covalent bonds
(D) Van der Waals forces
12. In a fetus, where are lymphocytes produced?
(A) In the spleen
(B) In the bone marrow
(C) In the liver
(D) In the heart
13. Anticodon is present in
(A) DNA
(B) tRNA
(C) rRNA
(D) mRNA
14. Name the secondary structure of tRNA?
(A) Cloverleaf
(B) L-shaped
(C) Duplex
(D) Triple Helix
15. Glycolysis is the conversion of
(A) Fructose into phosphoenolpyruvate
(B) Fructose into pyruvate
(C) Glucose into phosphoenolpyruvate
(D) Glucose into pyruvate
16. Which of the following rRNA molecules have peptidyl transferase activity in prokaryotes?
(A) 23S rRNA
(B) 28S rRNA

- (C) 5S Rna
- (D) 18S rRNA

17. The percentage of human genome which encodes proteins is approximately

- (A) Less than 2%
- (B) 5%
- (C) 25%
- (D) 99%

18. In EMP pathway, the process by which ATP is formed from ADP is

- (A) reduction
- (B) oxidative phosphorylation
- (C) substrate-level phosphorylation
- (D) photo phosphorylation

19. Sickle cell anemia is caused

- (A) When valine is replaced by glutamic acid in beta polypeptide chain
- (B) When glutamic acid is replaced by valine in beta polypeptide chain
- (C) When glutamic acid is replaced by valine in alpha polypeptide chain
- (D) When valine is replaced by glutamic acid in alpha polypeptide chain

20. In prokaryotic cells, ribosomes are

- (A) 70 S
- (B) 80 S
- (C) 60S + 40S
- (D) 50S + 40S

21. Which of these is a characteristic of prokaryotic cells?

- (A) Absence of cell organelles
- (B) Absence of nucleus
- (C) Presence of 70S ribosomes
- (D) All of these

22. In prokaryotes, the hair-like outgrowths which attach to the surface of other bacterial cells are

- (A) Flagella
- (B) Pili
- (C) Capsule
- (D) Plasmids

23. B-cells and T-cells are two types of cells involved in

- (A) Innate Immunity
- (B) Active immunity
- (C) Passive immunity
- (D) Acquired immunity

24. Who coined the term mitochondria?

- (A) Kolliker
- (B) Benda
- (C) Fleenming
- (D) Robert Brown

25. On the Origin of Species was written by

- (A) Charles Darwin
- (B) Ludmila Kuprianova
- (C) Mikhail A. Fedonkin
- (D) None of the above

26. Which of the following cells of the immune system do not perform phagocytosis?

- (A) Macrophage
- (B) Neutrophil
- (C) Eosinophil
- (D) Basophil

27. The force that initiates evolution is

- (A) Variation
- (B) Mutation
- (C) Extinction
- (D) Adaptation

28. The limbless amphibian is

- (A) *Icthyophis*
- (B) *Hyla*
- (C) *Rana*
- (D) *Salamander*

29. The EMP pathway in eukaryotes usually takes place in

- (A) nucleus
- (B) lysosome
- (C) Golgi apparatus
- (D) cytosol

30. What is the first step in the payoff phase of glycolysis?

- (A) Reduction of 1, 3-bisphosphoglycerate to glyceraldehyde 3-phosphate
- (B) Oxidation of glyceraldehyde 3-phosphate to 1, 3-bisphosphoglycerate
- (C) Reversible conversion of dihydroxyacetone phosphate to glyceraldehyde 3-phosphate
- (D) Irreversible conversion of dihydroxyacetone phosphate to glyceraldehyde 3-phosphate

31. Plasmamembrane is

- (A) Permeable
- (B) Selectively permeable
- (C) Impermeable
- (D) Semi-permeable

32. In mitochondria cristae act as sites for

- (A) proteinsynthesis
- (B) phosphorylation of flavoproteins
- (C) breakdown of macromolecules
- (D) Oxidation-reductionreaction

33. Which of the following statements is true regarding the "law of segregation"?

- (A) Law of segregation is the law of purity of genes
- (B) Alleles separate from each other during gametogenesis
- (C) Segregation of factors is due to the segregation of chromosomes during meiosis
- (D) All of the above

34. How many phenotypes can occur in the human blood group ABO with alleles $I^A I^B i$?

- (A) 2
- (B) 3
- (C) 4
- (D) 1

35. Cystic fibrosis is

- (A) Sex-linked recessive disorder
- (B) Autosomal dominant disorder
- (C) Autosomal recessive disorder
- (D) Sex-linked dominant disorder

36. Alleles are

- (A) Alternate forms of genes
- (B) Linked genes
- (C) Chromosomes that have crossed over
- (D) Homologous chromosomes

37. The experiment that simulated conditions thought to be present on the early earth

- (A) Hershey-Chase experiment
- (B) Geiger-Marsden experiment
- (C) Miller-Urey experiment
- (D) Schiehallion experiment

38. Observation of species on _____ heavily inspired Darwin's theory of evolution.

- (A) Ilha da Queimada Grande
- (B) Guatemala
- (C) Faroe Islands
- (D) Galapagos Islands

39. Mammal's heart is

- (A) Myogenic
- (B) neurogenic
- (C) voluntary
- (D) sympathetic

40. What is the full form of PPLO?

- (A) Pleural Parasite Like Organisms
- (B) Phosphatic Pneumonia Like Organisms
- (C) Pleuro Parasite Like Organisms
- (D) Pleuro Pneumonia Like Organisms

41. Which of the following cells is involved in humoral immunity?

- (A) T-cells
- (B) B-cells
- (C) Mast cells
- (D) Both T and B cells

42. Body cavity lined by mesoderm is called

- (A) coelenteron
- (B) pseudocoel
- (C) coelom
- (D) blastocoel

43. Which of the following components of the vertebrate immune response occurs first upon invasion by a virus or bacterium?

- (A) Activation of killer T lymphocytes
- (B) Activation of B lymphocytes
- (C) The inflammatory response
- (D) Mobilization of complement proteins

44. Tube feet are the locomotory organs of

- (A) platyhelminthes
- (B) Echinodermata
- (C) Mollusca
- (D) Arthropoda

45. Which of the following organelle is called 'Suicidal Bag'?

- (A) Mitochondria
- (B) Endoplasmicreticulum
- (C) Lysosome
- (D) Ribosome

B. Answer any 10 questions from the following.

2×10=20

1. What is pseudocoelom? Where is it found?
2. What is Nucleotide?
3. In which cell organelle cristae is found? Mention its one function.
4. Mention two differences between Adaptive Immunity and Innate Immunity.
5. Write down the two characteristics features of Phylum Mollusca.
6. What do you mean by EMP pathway?
7. What is Central Dogma in molecular biology?
8. Mention one prokaryotic cell and one eukaryotic cell name.
9. What is tRNA? Mention its one function.
10. What is Nephridia? Where it is found?
11. What do you mean by allele?
12. What is Codon?

C. Answer any 3 questions from the following. 5×3=15

1. Describe the structure of DNA with suitable diagram.
2. What is Glycolysis? Write down the steps of Glycolysis.
3. Write down the short note on B-cell and T-cell.
4. Write down the characteristics features of Phylum Echinodermata. Mention two animal name under this phylum.

ALIAH UNIVERSITY
Department of Civil Engineering

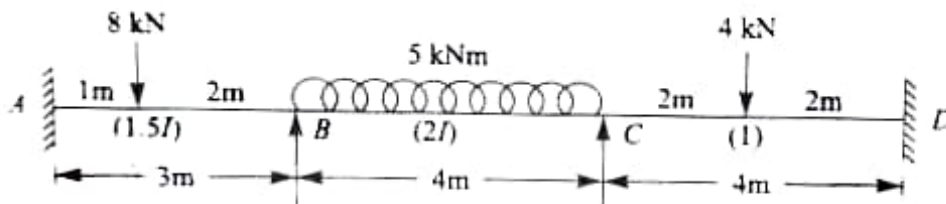
End Semester Examination(Spring), 2nd Year , 4th Semester
CENUGPC07 : Structural Analysis

Time – 3hrs

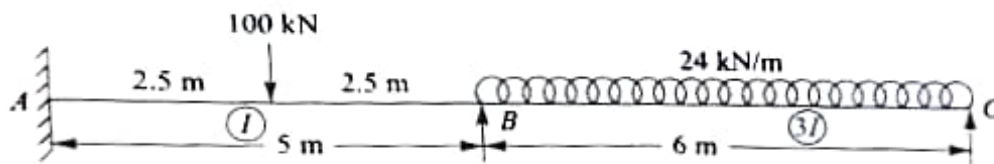
Total Marks – 80

(Answer All Questions)

1. Calculation the **Moment of Inertia** of the following:
 - (i) Rectangle about the C.G.
 - (ii) Triangle about its Base.
 - (iii) Circle about its Diameter
2. Draw the bending moment diagram using **moment distribution method** of the beam as shown below.



3. Find the reactions and also draw the bending moment diagram using **slope deflection method** for the beam as shown below.



4. Draw the bending moment and shear diagram for the beams as shown below.

